Evolved policies vs baseline — CORAL run 2026-10-09_004013

Parameters used in the score normalization

Probe	Weight	Direction	Baseline mean (noop, n=8)	Baseline sd	CV %	z	Contribution
Throughput (ops/s)	2.00	maximize	1,224.32	9.52	0.78	(value − mean) / sd	tanh(z)
Bytes read (cgroup rbytes)	0.50	minimize	318,859,000 (318.9 MB)	3,868,000	1.21	(value − mean) / sd	tanh(−z)
Refaults (cgroup memstat)	0.25	minimize	51,054.2	918.7	1.80	(value − mean) / sd	tanh(−z)

score = (2.0 × c_throughput + 0.5 × c_bytes + 0.25 × c_refaults) / 2.75, where c is the tanh-squashed contribution listed in the table above. Range [−1, 1]; 0 = same as the noop baseline. Baseline = 8 consecutive noop evaluations. The displayed values reproduce all 28 recorded scores (max deviation < 0.002).

Run

Item	Value
Attempts	28 total = 22 evolved policies (16 real, 6 --tune) + 6 attempts running unmodified reference-policy code (context only)
Distinct evolved code versions	18
Definition of 'evolved'	an attempt whose code differs from every unmodified reference policy (fifo/mru/s3fifo/sampling/get_scan/mglru/lhd) the agent submitted as-is
Best evolved policy	#17 61054f8b (score 0.9980); details at the end of this report

Unmodified reference policies — context only (not evolved)

#	Commit	Cls	Throughput (ops/s)				Bytes read (MB)				Refaults (count)				Score	Description
			value	Δ%	z	contrib	value	Δ%	z	contrib	value	Δ%	z	contrib		
1	3220c3bb	real	1,075.1	-12.2	-15.7	-1.00	270.2	-15.3	-12.6	+1.00	39,626	-22.4	-12.4	+1.00	-0.4545	baseline: reference get_scan.c unchanged (scan-pid list + sampled LFU w/ index-block bonus) to calib…
2	7c3ef53d	real	1,263.0	+3.2	+4.1	+1.00	314.9	-1.2	-1.0	+0.77	50,172	-1.7	-1.0	+0.74	0.9343	reference s3fifo.c unchanged (landscape mapping)
3	995ee7be	real	1,257.6	+2.7	+3.5	+1.00	317.0	-0.6	-0.5	+0.45	50,720	-0.7	-0.4	+0.35	0.8388	reference mglru.c unchanged (landscape mapping)
6	9fd454e5	real	1,261.5	+3.0	+3.9	+1.00	310.8	-2.5	-2.1	+0.97	49,226	-3.6	-2.0	+0.96	0.9906	noise check: plain s3fifo re-run (identical to 7c3ef53d except trailing comment)
19	ef625089	tune	1,141.3	-6.8	-8.7	-1.00	283.7	-11.0	-9.1	+1.00	42,743	-16.3	-9.1	+1.00	-0.4545	tune probe: reference sampling.c unchanged (landscape point + learn what --tune does on this grader)
25	df506481	tune	1,070.5	-12.6	-16.1	-1.00	278.9	-12.5	-10.3	+1.00	41,698	-18.3	-10.2	+1.00	-0.4545	landscape: reference lhd.c unchanged (last unmapped reference policy)

Attempt #6 re-ran the unmodified reference s3fifo code (comment-only difference from #2).

Evolved policies compared with the baseline

#	Commit	Cls	Throughput (ops/s)				Bytes read (MB)				Refaults (count)				Score	Description
			value	Δ%	z	contrib	value	Δ%	z	contrib	value	Δ%	z	contrib		
4	1ab2df0b	real	1,232.7	+0.7	+0.9	+0.70	323.3	+1.4	+1.2	-0.82	52,177	+2.2	+1.2	-0.84	0.2863	s3fifo + scan_pids: scan-thread accesses don't bump freq (sequential scans touch each page many time…
5	0e4bf02e	real	1,237.2	+1.0	+1.4	+0.87	316.9	-0.6	-0.5	+0.47	50,657	-0.8	-0.4	+0.41	0.7587	s3fifo + correlated-reference period: accesses within 2ms of the last counted one don't bump freq (f…
7	b96b2ce0	real	1,256.2	+2.6	+3.4	+1.00	309.1	-3.1	-2.5	+0.99	48,787	-4.4	-2.5	+0.99	0.9946	s3fifo with small queue 1/10 of cache instead of 1/15 (canonical S3-FIFO ratio)
8	4dfced51	real	1,216.4	-0.6	-0.8	-0.68	305.0	-4.3	-3.6	+1.00	47,831	-6.3	-3.5	+1.00	-0.2228	s3fifo small 1/10 + ghost map 2x cache size (remember more evicted pages so quick re-references go s…
9	a2f787dd	real	1,276.4	+4.3	+5.5	+1.00	311.8	-2.2	-1.8	+0.95	49,470	-3.1	-1.7	+0.94	0.9852	s3fifo 1/10 + ghost 2x, but scan-thread re-adds never promote via ghost (test: does scan ghost-flood…
10	b631bf1b	real	1,273.7	+4.0	+5.2	+1.00	311.1	-2.4	-2.0	+0.96	49,301	-3.4	-1.9	+0.96	0.9894	gated-ghost s3fifo with ghost 4x cache (only GET-thread re-adds use ghost, so remember more GET reus…
11	2d5d649a	real	1,277.7	+4.4	+5.6	+1.00	317.2	-0.5	-0.4	+0.41	50,729	-0.6	-0.3	+0.34	0.8321	gated-ghost s3fifo (ghost 4x) with small queue 1/8 of cache (bigger small queue cut I/O in eval 7; t…
12	777cbda4	real	1,237.8	+1.1	+1.4	+0.89	312.6	-2.0	-1.6	+0.93	49,628	-2.8	-1.6	+0.91	0.8980	gated-ghost s3fifo (4x, small 1/10): scan-thread ghost hits now enter main at the HEAD (low priority…
13	c14b63da	real	1,141.8	-6.7	-8.7	-1.00	283.0	-11.2	-9.3	+1.00	42,637	-16.5	-9.2	+1.00	-0.4545	structural attempt 1/3 on GET-aware aging: gated-ghost s3fifo (4x, 1/10) + get_touched bit; main CLO…
14	c01ca9a6	real	1,228.2	+0.3	+0.4	+0.39	324.7	+1.8	+1.5	-0.91	52,518	+2.9	+1.6	-0.92	0.0326	structural attempt 2/3 on GET-aware aging: milder penalty - scan-only pages freq cap 2 (GET-touched …
15	a50fe634	real	1,255.2	+2.5	+3.2	+1.00	310.0	-2.8	-2.3	+0.98	49,001	-4.0	-2.2	+0.98	0.9921	structural attempt 3/3 on GET-aware aging: cap-2 for scan-only pages, but get_touched set ONLY at ad…
16	5e23d6c7	real	1,247.4	+1.9	+2.4	+0.98	311.2	-2.4	-2.0	+0.96	49,286	-3.5	-1.9	+0.96	0.9781	hook-overhead attempt 1/3: gated-ghost s3fifo (b631bf1b) with cheaper folio_accessed - drop watchlis…
17	61054f8b	real	1,257.4	+2.7	+3.5	+1.00	305.6	-4.2	-3.4	+1.00	48,005	-6.0	-3.3	+1.00	0.9980	drift check: re-run gated-ghost s3fifo b631bf1b unchanged (eval16's cheaper hook gave 1247 vs 1274 o…
18	ce24b92d	real	1,258.8	+2.8	+3.6	+1.00	308.4	-3.3	-2.7	+0.99	48,673	-4.7	-2.6	+0.99	0.9963	gated-ghost s3fifo (b631bf1b) + LevelDB index-block protection: last page of each file promoted from…
20	80c98286	tune	1,273.1	+4.0	+5.1	+1.00	310.3	-2.7	-2.2	+0.98	49,109	-3.8	-2.1	+0.97	0.9930	variance check: 3rd run of gated-ghost s3fifo b631bf1b (runs so far: 1273.7/311.1MB, 1257.4/305.6MB)
21	23aef7a1	tune	1,211.4	-1.1	-1.4	-0.88	327.8	+2.8	+2.3	-0.98	53,273	+4.3	+2.4	-0.98	-0.9040	ablation: gated-ghost s3fifo (small 1/10) with ghost back to 1x cache - is the bigger ghost load-bea…
22	b7395753	real	1,272.9	+4.0	+5.1	+1.00	312.0	-2.1	-1.8	+0.94	49,530	-3.0	-1.7	+0.93	0.9832	gated-ghost s3fifo, ghost 2x, scan-added pages that never reached main get NO ghost entry (eval21 ab…
23	ff01f9d1	tune	1,254.3	+2.4	+3.1	+1.00	307.8	-3.5	-2.9	+0.99	48,503	-5.0	-2.8	+0.99	0.9954	b7395753 + GET-added pages promote small->main on first reuse (freq>0); scan-added pages still need …
24	60071e54	tune	1,244.7	+1.7	+2.1	+0.97	308.8	-3.2	-2.6	+0.99	48,773	-4.5	-2.5	+0.99	0.9769	gated-ghost s3fifo b631bf1b with ghost map as common (global) LRU instead of BPF_F_NO_COMMON_LRU: pe…
26	e573de75	tune	1,259.7	+2.9	+3.7	+1.00	308.4	-3.3	-2.7	+0.99	48,604	-4.8	-2.7	+0.99	0.9967	gated-ghost s3fifo (61054f8b) with freq cap 2 instead of 3: fewer CLOCK passes in main, faster turno…
27	50f29a63	tune	1,258.4	+2.8	+3.6	+1.00	305.4	-4.2	-3.5	+1.00	47,966	-6.0	-3.4	+1.00	0.9983	repeat #2 of freq-cap-2 gated-ghost s3fifo (e573de75: 1259.7 ops/s, 308.4 MB) to compare mean vs b63…
28	c3bd68ae	real	1,165.0	-4.8	-6.2	-1.00	308.2	-3.4	-2.8	+0.99	48,549	-4.9	-2.7	+0.99	-0.4568	real submission of freq-cap-2 gated-ghost s3fifo (2 tune runs: 0.9967 / 0.9983, mean 306.9 MB vs bas…

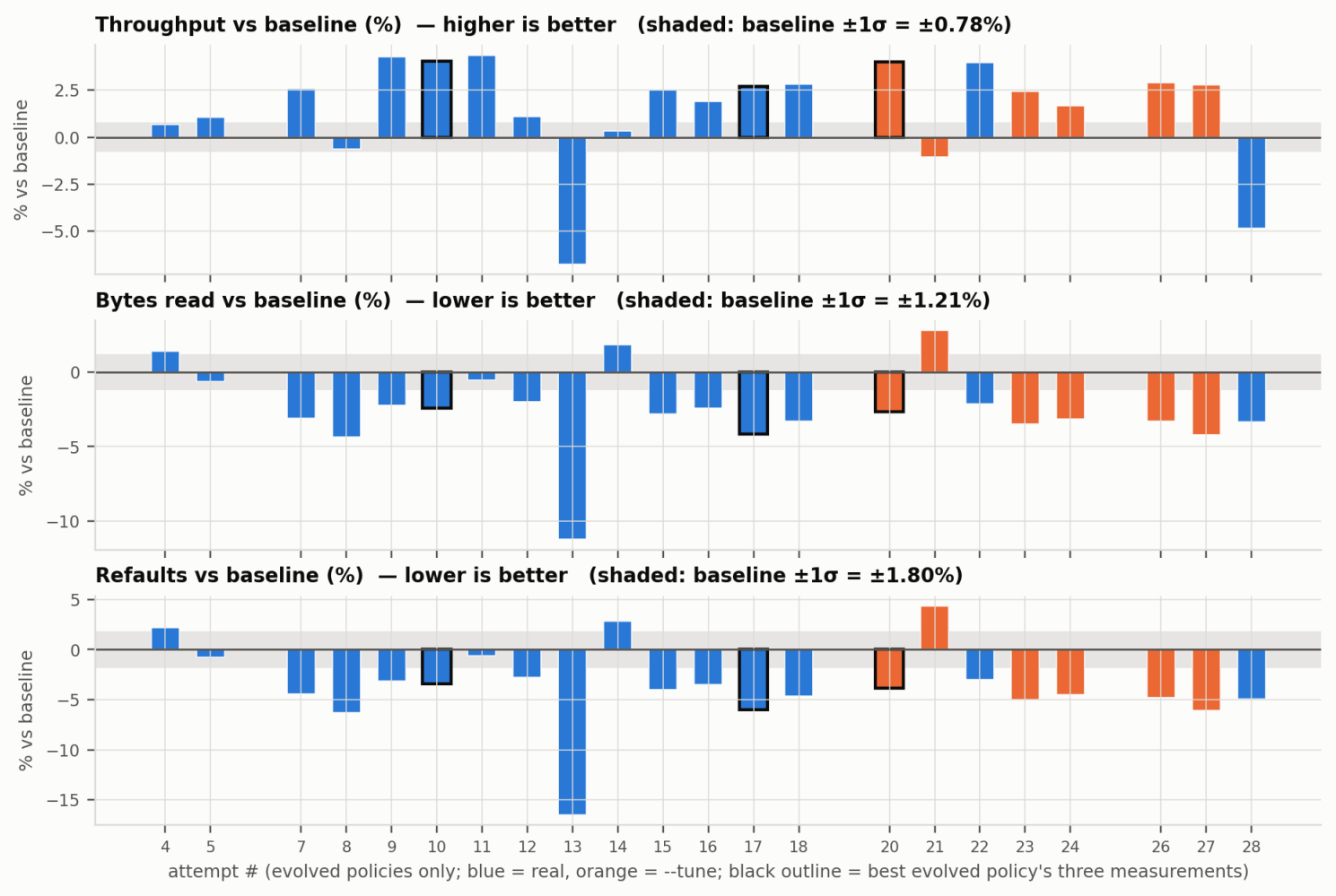
contrib = tanh(z) for throughput, tanh(−z) for bytes read and refaults. Baseline: throughput 1,224.3 ops/s, bytes read 318.9 MB, refaults 51,054. Δ% = (value − baseline mean) / baseline mean. Shaded rows = the best evolved policy's code (3 measurements).

Aggregate over the evolved policies

Probe	n	Mean Δ%	Median Δ%	Best Δ% (attempt)	Worst Δ% (attempt)	Better than baseline	Better by > 1σ	Better by > 2σ
Throughput	22	+1.49	+2.49	+4.36 (#11)	-6.74 (#13)	18 / 22	16	14
Bytes read	22	-2.51	-2.73	-11.24 (#13)	+2.80 (#21)	19 / 22	17	12
Refaults	22	-3.58	-3.92	-16.49 (#13)	+4.35 (#21)	19 / 22	17	12

Evolved policies better than baseline on all three probes at once: 16 of 22; by more than 2σ on all three: 9 of 22.

Difference from baseline, per evolved attempt



Best evolved policy — #17 61054f8b

Selection

Item	Value
Rule	highest score among the real (non --tune) evolved attempts, i.e. the top of CORAL's leaderboard once unmodified reference code is excluded
Selected attempt	#17 commit 61054f8b6c4d0ae99a774ab800e7957efe187adf score 0.9980
Identical code measured in	#10 b631bf1b (real, score 0.9894), #17 61054f8b (real, score 0.9980), #20 80c98286 (tune, score 0.9930)
Description (first attempt of this code)	gated-ghost s3fifo with ghost 4x cache (only GET-thread re-adds use ghost, so remember more GET reuse)
Source	report/best_policy.c (noop.c at commit 61054f8b; 629 lines)

Measurements of this policy vs baseline

Measurement	Throughput (ops/s)				Bytes read (MB)				Refaults				Score
	value	Δ%	z	contrib	value	Δ%	z	contrib	value	Δ%	z	contrib	
Baseline (noop mean ± sd)	1,224.3 ± 9.5				318.9 ± 3.87				51,054 ± 919				
#10 b631bf1b (real)	1,273.7	+4.03	+5.19	+1.000	311.1	-2.42	-1.99	+0.964	49,301	-3.43	-1.91	+0.957	0.9894
#17 61054f8b (real)	1,257.4	+2.70	+3.48	+0.998	305.6	-4.17	-3.43	+0.998	48,005	-5.97	-3.32	+0.997	0.9980
#20 80c98286 (tune)	1,273.1	+3.99	+5.12	+1.000	310.3	-2.67	-2.20	+0.976	49,109	-3.81	-2.12	+0.971	0.9930
Mean of 3 measurements	**1,268.1**	**+3.57**	**+4.60**	**+1.000**	**309.0**	**-3.09**	**-2.54**	**+0.988**	**48,805**	**-4.41**	**-2.45**	**+0.985**	**0.9963**
Sample sd of 3 measurements	9.2				3.01				699				0.0043

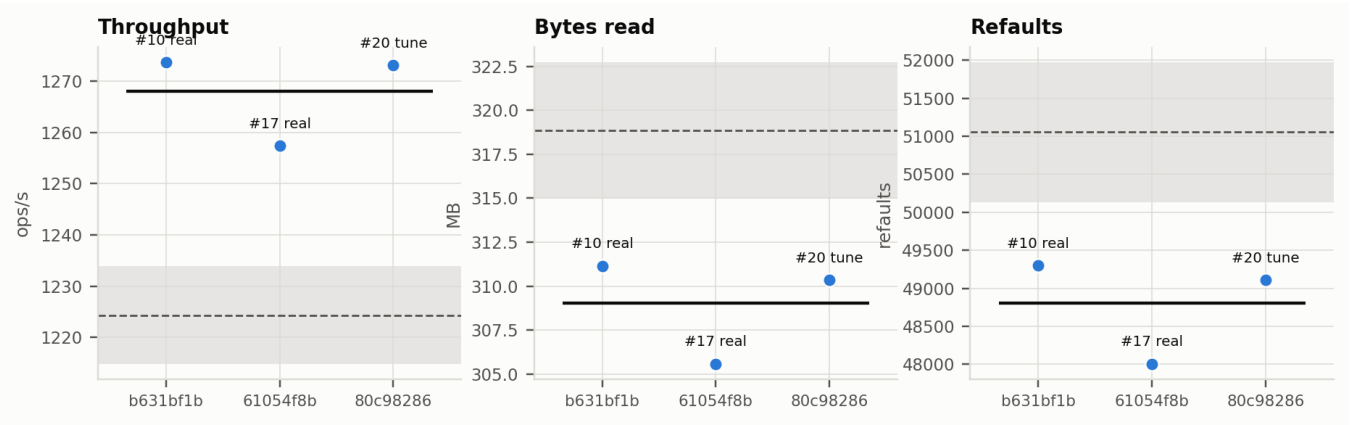
The mean row applies the same normalization (z vs baseline, tanh, weights) to the mean of the measurements.

Alternatives considered (evolved policies)

Basis	Attempt	Cls	Score	Throughput	Δ% tput	Δ% bytes	Δ% refaults	Identical measurements	Mean tput of that code	Best policy
top-5 by score (real)	#17 61054f8b	real	0.9980	1,257.4	+2.70	-4.2	-6.0	3	1,268.1	★
	#18 ce24b92d	real	0.9963	1,258.8	+2.81	-3.3	-4.7	1	1,258.8	
	#7 b96b2ce0	real	0.9946	1,256.2	+2.60	-3.1	-4.4	1	1,256.2	
	#15 a50fe634	real	0.9921	1,255.2	+2.53	-2.8	-4.0	1	1,255.2	
	#10 b631bf1b	real	0.9894	1,273.7	+4.03	-2.4	-3.4	3	1,268.1	★
top-3 by throughput (all)	#11 2d5d649a	real	0.8321	1,277.7	+4.36	-0.5	-0.6	1	1,277.7	
	#9 a2f787dd	real	0.9852	1,276.4	+4.25	-2.2	-3.1	1	1,276.4	
	#10 b631bf1b	real	0.9894	1,273.7	+4.03	-2.4	-3.4	3	1,268.1	★

The highest single throughput (#11 2d5d649a) is one measurement of a different code version; the repeat measurements of other versions show single-evaluation spread of several ops/s, so it is not separable from the selected policy.

Best evolved policy — measurements and changes from the reference



Changes relative to the reference s3fifo.c it was derived from — kernel: +12 / −2 lines, loader: +11 / −1 lines (code lines only; comments and blank lines ignored)

Section	±	Changed line
kernel	+	struct {
kernel	+	__uint(type, BPF_MAP_TYPE_HASH);
kernel	+	__type(key, int);
kernel	+	__type(value, bool);
kernel	+	__uint(max_entries, 100);
kernel	+	} scan_pids SEC(".maps");
kernel	+	static inline bool is_scanning_pid(void) {
kernel	+	int pid = (int)(bpf_get_current_pid_tgid() & 0xFFFFFFFF);
kernel	+	return bpf_map_lookup_elem(&scan_pids, &pid) != NULL;
kernel	+	}
kernel	−	if (small_list_size >= cache_size / 15 || main_list_size <= 2 * small_list_size)
kernel	+	if (small_list_size >= cache_size / 10 || main_list_size <= 2 * small_list_size)
kernel	−	if (folio_in_ghost(folio)) {
kernel	+	if (!is_scanning_pid() && folio_in_ghost(folio)) {
loader	−	if (bpf_map__set_max_entries(skel->maps.ghost_map, skel->rodata->cache_size)) {
loader	+	if (bpf_map__set_max_entries(skel->maps.ghost_map, 4 * skel->rodata->cache_size)) {
loader	+	{
loader	+	int scan_pids_fd = bpf_obj_get("/sys/fs/bpf/cache_ext/scan_pids");
loader	+	if (scan_pids_fd >= 0) {
loader	+	if (bpf_map__reuse_fd(skel->maps.scan_pids, scan_pids_fd))
loader	+	fprintf(stderr, "Failed to reuse scan_pids map fd\n");
loader	+	close(scan_pids_fd);
loader	+	} else {
loader	+	fprintf(stderr, "no scan_pids pin found\n");
loader	+	}
loader	+	}